

# Calculating the uncertainty in the width of a steel strip

IR18 – Industrial Statistics

Based on "A Beginner's Guide to Uncertainty" by Stéphanie Bell.

In the Cold Rolling Mill department of a steel plant, at the entrance of the galvanizing line, the width of the steel coils is measured using a tape measure. As an illustration we will calculate the uncertainty associated with the obtained result.

## Consider the sources of measurement uncertainty.

A SWIPE model could be used to identify the sources of measurement variation. In this case possible sources are :

### *(due to the tape measure)*

- What is the uncertainty in the calibration ? (\*)
- Is it prone to stretching ?
- Could bending have shortened it ?
- What is the resolution, i.e. how small are the divisions on the tape (e.g. millimetres) ? (\*)

### *(due to the measured coil)*

- Does the prevailing temperature affect its width ?
- Are the side ends well-shaped, or rather tapered ?
- Does the flatness affect the result ?

### *(due to the measuring process)*

- How well can the beginning of the tape be lined up with the side of the coil ?
- How well can the tape be laid at right angles to the coil ? (\*)
- How repeatable is the measurement ? (\*)

## Measurements.

Let us assume that each coil is measured 3 times<sup>1</sup>, aligning the tape measure freshly each time. The results are recorded to the nearest mm. For a given coil the average of the 3 measurements is 1834 mm, the range is 2 mm.

## Estimate the uncertainty of each input quantity that feeds into the final result.

Suppose that from the listed sources those indicated with a (\*) need to be considered<sup>2</sup>. For each of them the standard uncertainty  $u$  will be calculated.

- The calibration certificate for the tape measure learns that the calibration uncertainty is 0.1 % of reading at a coverage factor 2 (for a normal distribution). Therefore the standard uncertainty for this source is :  $u_1 = 1834 \times 0.1 \% \div 2 = 0.917$  mm.
- The divisions on the tape are millimetres. Reading to the nearest division gives an error of no more than  $\pm 0.5$  mm. We can take this to be a uniformly distributed uncertainty : the true width could lie anywhere in the 1 mm interval. The standard uncertainty  $u_2$  of this error source is equal to  $0.5 / \sqrt{3} = 0.289$  mm. (Proof this as an exercise.<sup>3</sup>)
- The tape lies straight but not always perpendicular to the coil. Therefore the measurement is likely to overestimate the actual width of the coil. Let us assume that the overestimate is about 0.1 % and that the uncertainty in this is also 0.1 %. This means that we should correct the result by subtracting 1.8 mm. The uncertainty  $u_3$  can be assumed to be uniformly distributed<sup>4</sup>. This means  $u_3 = 0.917 / \sqrt{3} = 0.529$  mm.
- The standard deviation of the measurements tells us about how repeatable the placement of the tape measure is, and how much it contributes to the uncertainty of the mean value. An estimate of the repeatability, based on the range<sup>5</sup> is  $2 / d_2 = 2 / 1.693$ . Therefore  $u_4 = 0.682$ .

Note that the first three estimates are type B estimates, while the last one is type A.

## Find the combined standard uncertainty $u_c$ .

Decide whether the errors are all independent. In this case let us say they are, although the repeatability may be influenced by the none perpendicular position of the tape measure. For simplicity we assume independency. This means that the complete error is the sum of the four errors mentioned above. The combined standard uncertainty can therefore be calculated as :

<sup>1</sup> There is a clear advantage of measuring three times. If one measures only once, and a mistake is being made leading to an erroneous result, this mistake will probably remain unnoticed. If one measures twice and the results are not in accordance, one knows that at least one of the measurements is not correct. By measuring three times there is a good reason to be suspicious about the measurement that does not agree with the other two.

<sup>2</sup> In reality other factors would probably need to be included.

<sup>3</sup> Hint : calculate the standard deviation of a uniformly distributed continuous random variable.

<sup>4</sup> Due to the fact that the angle in positioning the tape can be assumed to be uniformly distributed around 0°.

<sup>5</sup>  $E(R) = d_2\sigma$ .

$$u_c = \sqrt{0.917^2 + 0.289^2 + 0.529^2 + 0.682^2} = 1.264 .$$

Write down the result.

To get a confidence level of about 95 %, the combined standard uncertainty must be multiplied by a coverage factor of 2. Therefore with 95 % confidence, we conclude that the actual width of the coil is :

$$1834 - 1.8 \pm 2.5 \text{ mm} = 1832.2 \pm 2.5 \text{ mm} = [1829.7 ; 1834.7] \text{ mm}.$$